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JOINT IMF/WORLD BANK DEBT SUSTAINABILITY ANALYSIS 2013

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Tonga's rating has been changed from high to moderate risk of debt distress based on the joint IMF-WB low-income country debt sustainability assessment (LIC DSA). This change is mainly due to the improvement of the Country Policy and Institutional Assessment (CPIA) rating from 'Weak' to 'Medium' by the World Bank last year. Although the relevant indicators of external and public debt and debt service show a similar path for debt dynamics as last year's DSA, the upgrade of the CPIA rating for Tonga significantly increases thresholds of these indicators, with only a single threshold (PV of debt in percent of exports) now breached for no more than three years under the baseline scenario. Because remittances have declined significantly in recent years, they are excluded from debt burden thresholds and indicators under the baseline scenario, but may mitigate liquidity risks. Notwithstanding the upgrade of the DSA rating, risks remain signficant, and a cautious approach is recommended towards new nonconcessional loans. With scheduled repayments on two loans (reconstruction and roads) from the China EXIM Bank, Tonga's debt service burden is expected to rise sharply from FY 2013/14 (July 1 to June 30), substantially draining the government's cash balance. The government's balance sheet is also exposed to a currency risk, with about 60 percent of the outstanding external debt denominated in Chinese renminbi. A declining path of external public debt is critically dependent on steadfast implementation of medium-term fiscal consolidation, as outlined in the Tonga Strategic Development Framework (TSDF).

BACKGROUND

1. As a small open economy, Tonga is facing the dual challenge of large fluctuations on the external side and a limited production base on the domestic side. On one hand, it heavily relies on remittances, tourism, and foreign aid. On the other hand, the production and export bases are quite narrow and based mainly on a few agricultural products and fishery.

2. The economy is gradually recovering from the effects of the global slowdown and the domestic credit crunch. After averaging about 3 percent during the previous three years, economic growth slowed down to 0.8 percent in FY 2011/12. In the previous three years, public investment projects, particularly those funded by China EXIM Bank, had offset spillovers from the global financial crisis. However, those projects have been completed, and a normalization of construction activity has reduced the impetus for growth—along with weaknesses in remittances, other investment spending, and tourist arrivals. In FY 2012/13, the economy is expected to grow by 0.5 percent. Starting in FY 2013/14, however, a gradual recovery of remittances and tourism—along with improved infrastructure—will help growth to gradually increase to about 1¾ percent over the medium term.

3. Tonga's total public sector debt stock (including publicly guaranteed debt) has risen since FY 2008/09 and is expected to reach about 45 percent of GDP by the end of FY 2012/13. External debt accounts for the majority of the debt stock, amounting to about 40 percent of GDP, leaving Tonga vulnerable to foreign exchange liquidity and currency risks. This rise in debt stock was primarily on account of two loans (for reconstruction and roads) from China EXIM Bank[1]. These loans account for about 60 percent of the outstanding external debt and pose an excessive exposure to a single foreign currency, while long-term loans by international financial institutions (IFIs) account for about 35 percent of external debt. In the current fiscal year, disbursements by China EXIM Bank have been completed for both the reconstruction and road loans. In total, about 80 percent of the reconstruction loan was used to finance direct government capital expenditure while the remainder was used for on-lending to the private sector for office, residential, and retail construction. The terms of the on-lending include an interest of 5 percent on the outstanding Tongan pa'anga amount, and the government receives repayments from the private borrowers after the completion of the project[2]. The entire road reconstruction loan was used to finance direct government capital spending on road improvements, including on the outer islands.

[1]These loan commitments were signed in November 2007 (for reconstruction), and in February 2010 (for roads), and bear a 2 percent interest rate, 5-year grace period, and 20-year maturity. The arrangement with the Chinese contractor performing the work associated with these loans required an initial disbursement of 30 percent to cover the down payment, while outstanding disbursements were paid quarterly as work was completed and claims filed. The authorities hope to negotiate with China for a debt relief or a debt rescheduling for another 5 to 10 year grace period.

[2] Since March 2012, the private borrower has started to repay the government.

4. Tonga's DSA builds on the baseline scenario assumptions presented in Box 1. Under the baseline scenario, over the medium term, recovery of remittances and tourism, along with improved infrastructure will help growth to gradually increase to about 1¾ percent, slightly lower than its long-term average growth rate of about 2 percent. Public spending is assumed to grow slower than projected revenues and grants over the medium term, limiting the longer-term need for additional external borrowing. Remittances—the largest source of foreign exchange earnings (one-half of Tongans live abroad, mostly in Australia, New Zealand, and the United States)—are assumed to recover in the medium to long term, in line with economic recovery in remittance-originating countries, and remain an important source of foreign exchange inflows. The recovery in remittances is, however, more sluggish than assumed in the previous DSA. Remittances are projected to recover from 14 percent of GDP in FY 2011/12 to 17 percent of GDP by FY 2017/18 and an average of 18 percent of GDP over the longer term. This level is roughly the average during 1994–2000, but much lower than the 32 percent of GDP average observed during 2001–10. Because of recent declines, debt burden indicators and thresholds are exclusive of remittances under the baseline scenario.

5. On the fiscal front, the government is expected to implement a medium-term fiscal reform plan supported by tax reform and restraint on payroll expenditure while increasing priority spending. Under these assumptions, the government will be able to rebuild cash balances and maintain a declining debt path while covering rising debt service costs. Revenues are expected to get some boost from proposed reforms to tax policy and administration, as well as the gradual economic recovery, and this will hep offset the gradual decrease of cash grants. The government is expected to continue with its policy to reduce the share of the wage bill to 45 percent of recurrent spending, bringing down current expenditure by about 2 percent of GDP in the medium term. Given all of these, the medium-term overall fiscal balance (excluding the China EXIM Bank loans) is projected to average a surplus of about 1–2 percent of GDP.

6. The Country Policy and Institutional Assessment (CPIA) rating for Tonga was upgraded from 'Weak' to 'Medium' by the World Bank last year. Credit was given for major gains in macroeconomic policy management, social protection systems, and business environment reforms. Consistent with the stronger rating for Tonga's macroeconomic policies and institutions, it's indicative debt distress thresholds were raised relative to 2012 levels, thereby improving the country's performance under baseline and alternative debt scenarios (Figures 1a-b).

7. Based on the updated DSA, Tonga has been upgraded from high to moderate risk of external debt distress[3]. Only one threshold (PV of debt in percent of export) is still breached under the baseline scenario, and this breach is not protracted, lasting for only three years.

Thresholds for External Debt
(In percent)

	Threshold	
Indicator	Before	Now
Present value of debt to GDP	30	40
Present value of debt to exports	100	150
Present value of debt to revenue	200	250
Debt service to exports	15	20
Debt service to revenue	25	20

[3] This judgment reflects the *Staff Guidance Note on the Application of the Joint Fund-Bank Debt Sustainability Framework for Low-Income Countries* and *Revisiting the Debt Sustainability Framework for Low-Income Countries*.

While remittances are excluded from debt burden indicators and thresholds, remittances are expected to help mitigate liquidity risk.[4]

8. However, Tonga continues to face important debt vulnerabilities, and the current DSA suggests a similar path for debt dynamics as the previous year's DSA. The debt burden peaks early in the projection period and gradually declines in the absence of new nonconcessional external borrowing. It also assumes that going forward, the Tongan government will undertake planned revenue reforms which, together with restraints on recurrent spending, will contribute to a buildup of adequate cash reserves. It also assumes that the Tongan government will continue to refrain from taking new nonconcessional loans until Tonga is upgraded to a low risk of debt distress classification. It does not factor in any debt relief or rescheduling.

EXTERNAL DSA

Baseline

9. Under the baseline, due to the upgrade in the CPIA rating, the external debt trajectory currently breaches only one policy-dependent threshold, the PV of debt-to-export ratio. However, this breach is not protracted, with the ratio falling below the threshold in FY 2015/16 and remaining at lower levels over the medium term (Figure 1a)[5]. The PV of public and publicly guaranteed (PPG) external debt is currently about 35 percent of GDP, which breaches the former indicative threshold (30 percent of GDP) but is below the revised indicative threshold (40 percent of GDP). The PV of debt-to-revenue ratio, meanwhile, would stay well below the threshold throughout the projection period.

10. Debt service is expected to rise in the medium term on account of the expiration of the grace period on the China EXIM Bank loans. But it will remain well below the threshold of both exports and revenue (Figure 1a).

11. Tonga's remittances would help reduce liquidity risks. Remittances to Tonga have averaged about twice export receipts over the past decade, and have provided a considerable source of foreign exchange inflows. Because they have declined significantly in recent years, remittances are excluded from debt burden thresholds and indicators under the baseline scenario. In an alternative scenario including remittances, the PV of PPG external debt relative to exports and remittance is projected to stay below the

[4] T*he Interim Guidance on the Use of Remittances in the DSA Framework for Low-income Countries*, which considers remittances as part of the base case for countries with large remittances, will be used from next year.

[5] According to the DSA guidelines, countries should be considered at a high risk of debt distress if there is a protracted breach of debt or debt-service thresholds under the baseline scenario.

(lower) modified threshold of 120 percent[6][7]. When remittances are considered with export earnings, external debt service remains well under 10 percent throughout the projection period (Figure 1b).

Sensitivity Analysis

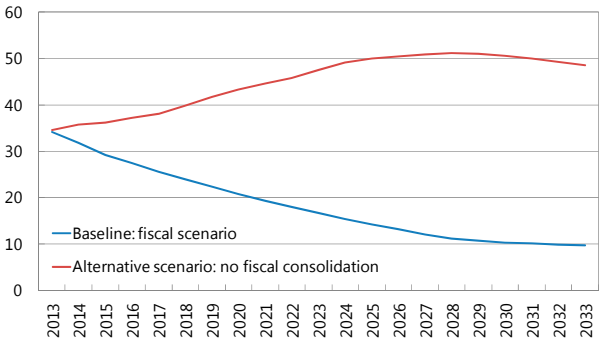
12. Risk of debt distress is sensitive to assumptions regarding the external environment. Stress tests show that the debt position is vulnerable to a significant slowdown in exports and currency risks. An export shock stress test causes the PV of debt-to-export ratio to approach 260 percent of exports and remain above the indicative threshold in the medium terms. Likewise, an exchange rate depreciation shock causes the PV of debt-to-GDP ratio to breach the indicative threshold until FY 2015/16. The alternative scenario based on historical averages (red lines) provides a more sanguine picture than the baseline[8], mainly because inflation rates of the U.S dollar denominated GDP deflator was higher in the past.

PUBLIC SECTOR DSA

Baseline

13. The public sector DSA reinforces the conclusions of the external DSA[9]. The high level of public sector borrowing for road building and reconstruction of the capital city over the past few years pushed PV of debt to stand to 40 percent of GDP in the current fiscal year. But it is expected to decline steadily under the baseline scenario (Figure 2). This highlights the importance of fiscal prudence and commitment to limiting new nonconcessional PPG borrowing[10].

NPV of Public Sector Debt Under Different Scenarios
(In percent of GDP)



Source: Fund staff calculations.

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[6] Under recently-issued guidance (Interim Guidance on the Use of Remittances), remittances will be included in the base case starting with the 2014 DSA. This is likely to reinforce the conclusion that Tonga is at moderate risk of debt distress.

[7] In a recent review, the IMF Executive Board approved modifications in some of the thresholds.

[8] In this alternative scenario, the following four variables are set at their historical levels; (i) real GDP growth; (ii) growth of GDP deflator (in U.S. dollar terms); (iii) non-interest current account in percent of GDP; and (iv) non-debt creating flows.

[9] The public sector DSA covers all Government and Government-guaranteed debt. Public enterprises are not included but all external borrowing by them is accounted for, as public enterprises do not have access to international capital markets.

10 Domestic debt (including government guaranteed debt) reached $36.9 million in FY 2012 (about 4.6 percent of GDP), comprising of GoT's bonds issued, with the majority of holdings by Financial Institutions and Retirement Fund Board. GoT's bonds are mainly issued on a roll-over basis with the exception of any redemptions or new issues.

Sensitivity Analysis and Alternative Scenario

14. Stress tests indicate that vulnerabilities remain throughout the projection period, especially to a significant depreciation (Figure 2). The PV of public debt would increase up to 53 percent in FY 2013/14 when a one-time 30 percent real depreciation takes places, but subsequently fall to 42 percent of GDP over the medium term. It should settle at around 19 percent of GDP in FY 2032/33.

15. Alternative scenarios highlight the importance of fiscal consolidation and growth enhancing reforms. An alternative scenario that maintains the primary balance at the FY 2011/12 level results in a slightly slower decline in debt and debt service indicators relative to the baseline. The other alternative scenario, where the long-term real GDP growth and primary balance are fixed at the average of the past decade (both of which are significantly worse than in the baseline), leads the relevant debt and debt service indicators either to fail to improve (PV of debt to GDP ratio and PV of debt to revenue ratio) or to improve only marginally (debt service to revenue ratio).

16. A no-reform scenario also highlights the importance of following through with the government's fiscal reform strategy. In a scenario of no tax reforms coupled with a failure to cut wages expenditures (including due to socio-economical challenges), the authorities will have to borrow domestically to bridge the financing gap. This will cause substantial further stress on the public debt level, increasing the PV of the debt-to-GDP ratio to about 50 percent over time.

STAFF ASSESSMENT

17. Tonga has moved from high to moderate risk of debt distress. This change is driven by an improvement in Tonga's CPIA rating from 'weak' to 'medium' and associated changes in debt distress thresholds. While not included in debt burden indicators and thresholds under the baseline scenario, remittances will mitigate liquidity risks. Threshold breaches under alternative scenarios indicate continued external vulnerabilities.

18. The assessment of reduced risk of debt distress is heavily contingent on successful fiscal consolidation. The fiscal authorities need to build enough cash balances (about 2 months of recurrent expenditure) to ensure servicing of the increased external debt obligations from FY 2013/14 without sacrificing essential public spending[11]. Alternative scenarios underline the importance of sustained fiscal consolidation for effective debt management. Additionally, there is a currency mismatch on the government's balance sheet (about 60 percent of the external debt are denominated in Chinese renminbi) which, if left unhedged, could pose additional risk to Tonga's debt dynamics.

19. Key medium-term vulnerabilities include major external shocks, currency mismatches, and lower GDP growth. These vulnerabilities underscore the importance of sound macroeconomic policies to improve Tonga's growth potential on a sustainable basis (as highlighted in the staff report), diversification

[11] More detailed information is provided in the staff report.

of export products and markets, and continued efforts to build fiscal buffers. Sound public debt management is also essential to guide future development financing in Tonga. Priority should be given to projects that would generate higher growth and employment, especially in the context of the Tonga Strategic Development Framework (TSDF), to help strengthen debt service capacity in the future. Consideration should also be given to hedging exchange risks on foreign liabilities using financial instruments, and developing a comprehensive debt management strategy with technical assistance.

The Authorities' Views

20. The authorities recognize the risks posed by the high debt burden and would like to bring down the public debt below 40 percent of GDP. The authorities agreed that creating fiscal space will be crucial in reducing the debt burden and protecting priority spending. In discussions with staff, the authorities noted that they were seeking to reschedule some of Tonga's debt obligations to China EXIM Bank. They shared staff's concerns about the currency risk posed by the current external debt structure, and noted that they were seeking technical assistance to mitigate this risk.

Box 1. Key Macroeconomic Assumptions

- **Real GDP growth** is projected at 0.5 percent in FY 2012/13. In the previous three years, spillovers from the global economic crisis were offset by public investments, in particular reconstruction and road projects funded by China EXIM Bank in FY 2011/12. However, as construction activities normalize and the recovery of remittances remains slow, economic growth is expected to be weak in FY 2012/13. Over the medium term, recovery of remittances and tourism, along with improved infrastructure, will help growth gradually increase to about 1¾ percent, slightly lower than its long term average growth rate of about 2 percent.

- The government is assumed to achieve an **overall fiscal surplus** of about 1−2 percent of GDP over the medium term, including to meet the expected debt repayment. Reforms in tax policy and administration are expected to improve the tax-to-GDP ratio to 16.4 percent of GDP from 15.7 percent of GDP over the medium term, while cash grants will decline to 4.6 percent of GDP in FY 2017/18 from 9.9 percent of GDP in FY 2011/12. Current expenditures (excluding ones funded by project grants) are expected to slow down, led by a reduction in the ratio of the wage bill to GDP by about 1 percentage points over the medium term. Total capital expenditure is expected to increase from about 2 percent of GDP to 3.4 percent of GDP over the medium term.

- **Over the longer term,** the primary fiscal surplus is expected to be eliminated with the gradual reduction in cash grants. Lower project grants primarily explain the reduction in expenditure, while the wage bill is targeted to average 45 percent of domestically-funded expenditure. Total capital expenditure is assumed to remain at about 4 percent of GDP.

- **Financing**. Majority of the grants, together with projected disbursements from concessional loans, are assumed to be spent on development projects and associated maintenance. Following the final disbursement of the China EXIM Bank road loan, there will be no external nonconcessional borrowing until Tonga is upgraded to low risk of debt distress. The external financing from the World Bank and Asian Development is expected to be under the rule of 50:50 (grants vs loans). On the domestic front, no new (net) domestic borrowing is assumed.

- The **current account deficit** is projected to narrow from 6.6 percent of GDP in FY 2011/12 to 5.9 percent of GDP in FY 2012/13, due mainly to a decline in imports. The current account deficit should average about 3 percent of GDP over the medium term and narrow slightly over the long term, as remittances and tourism receipts improve.

- **Remittances** are expected to recover during the current fiscal year, along with strengthening economic prospects in Australia, New Zealand, and the United States. Remittances are expected to remain an important source of foreign exchange in the medium term.

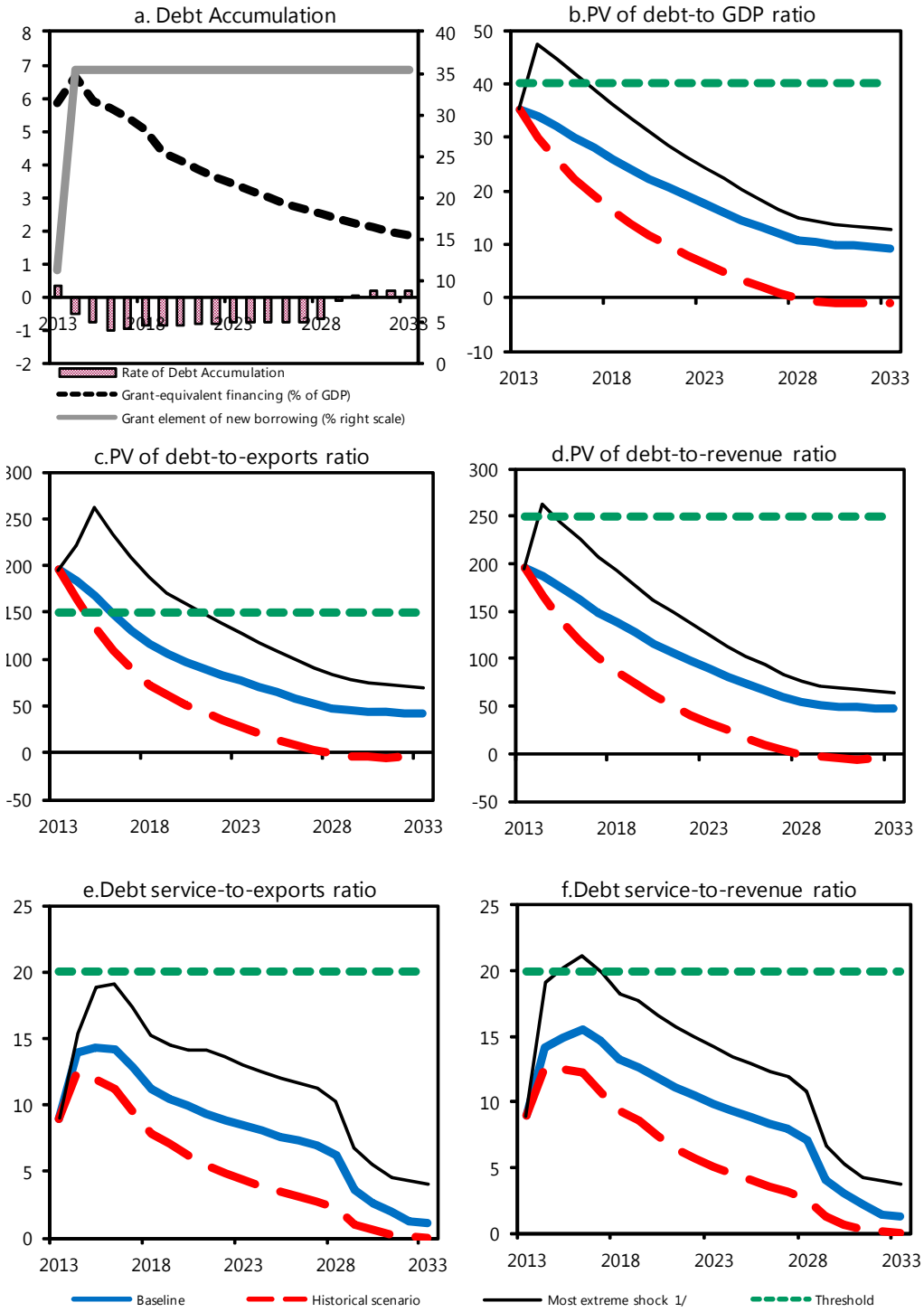
- **Tourism** receipts are projected to average around 10 percent of GDP and increase to an average of 13 percent of GDP over the long term assuming that the tourism environment continues to improve.

- The **export** base is projected to remain narrow and relatively undiversified. The proper operation of the fumigation facility should help exports by meeting New Zealand's bio-security requirements.

Tonga: Key Macroeconomic Assumptions (in percent)

	2012 DSA	2013 DSA
	2012-17 AVG	2013-18 AVG
Real GDP growth	1.7	1.5
GDP deflator in US dollar terms (change)	3.8	1.5
Effective interest rate	1.6	1.9
Growth of exports of G&S (US dollar terms)	7.2	7.6
Growth of imports of G&S (US dollar terms)	6.6	2.9
Grant element of new public sector borrowing	34.9	31.3

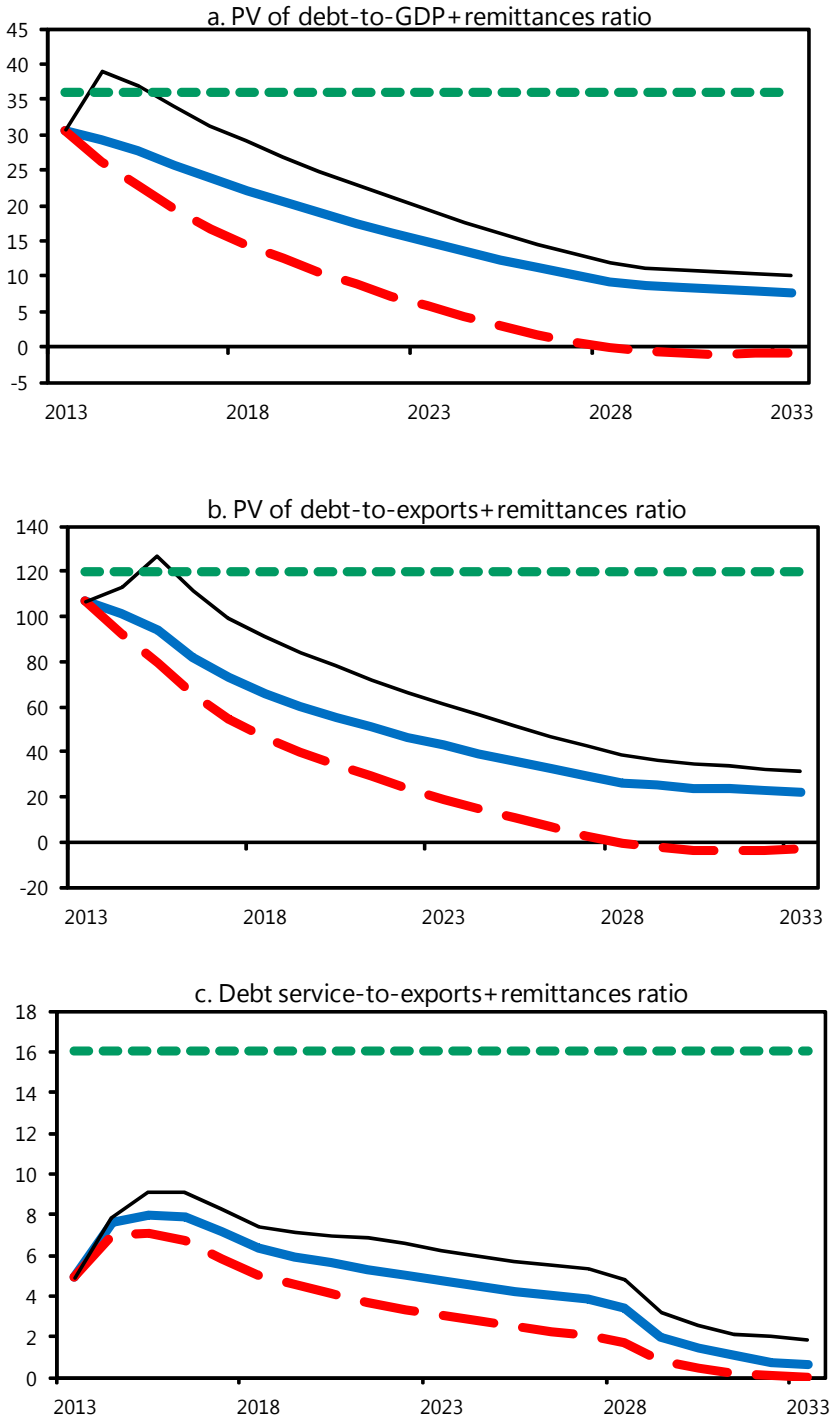
Figure 1a. Tonga: Indicators of Public and Publicly Guaranteed External Debt under Alternative Scenarios, 2013–33 1/



Sources: Country authorities; and staff estimates and projections.

1/ The most extreme stress test is the test that yields the highest ratio in 2023. In figure b. it corresponds to a one-time depreciation shock; in c. to a exports shock; in d. to a one-time depreciation shock; in e. to a exports shock and in figure f. to a one-time depreciation shock.

Figure 1b. Tonga: Indicators of Public and Publicly Guaranteed External Debt under Alternative Scenarios, with Remittances, 2013–33 1/



Sources: Country authorities; and staff estimates and projections.

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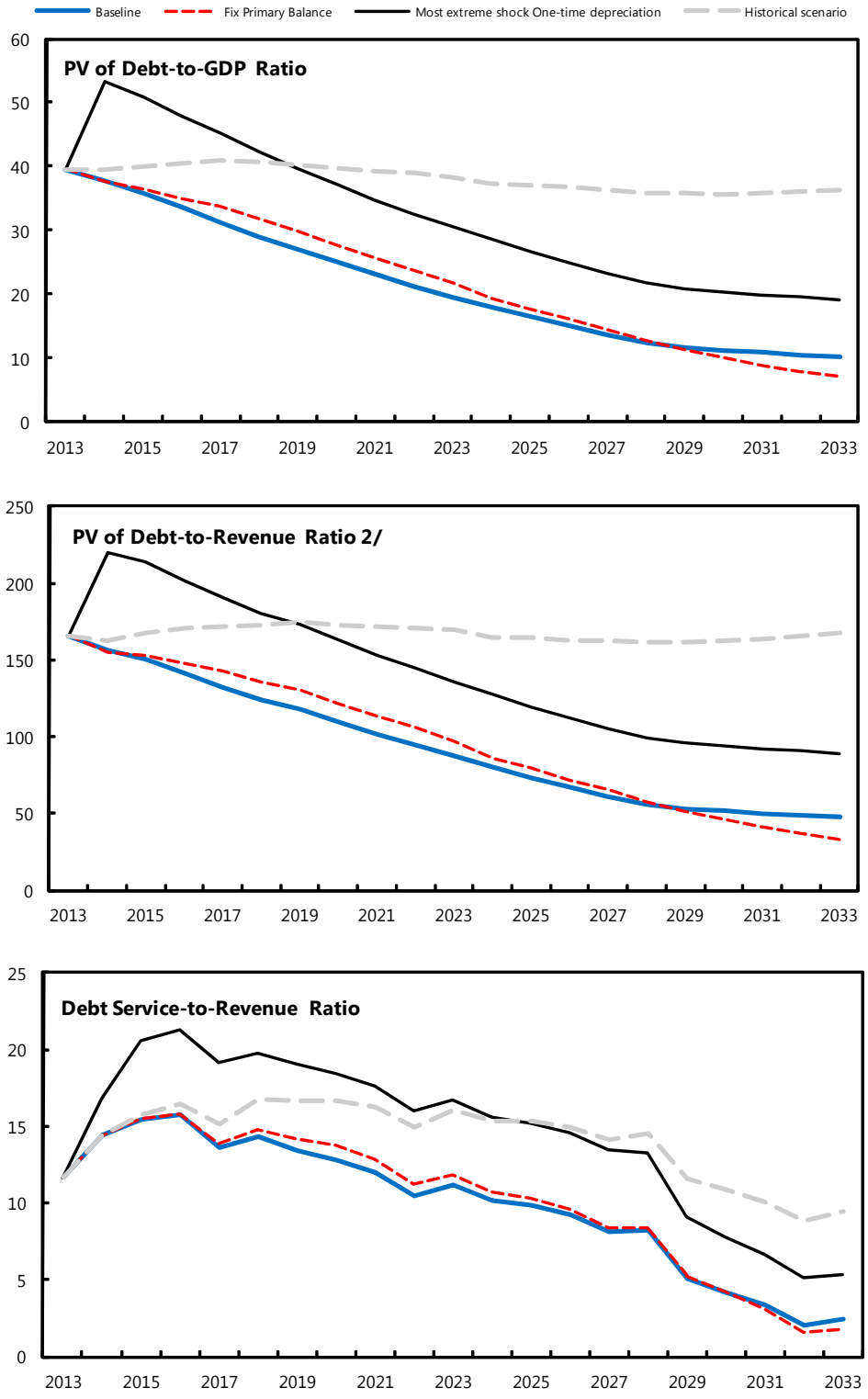
Figure 2. Tonga: Indicators of Public Debt under Alternative Scenarios, 2013–33 1/



Sources: Country authorities; and staff estimates and projections.
1/ The most extreme stress test is the test that yields the highest ratio in 2023.
2/ Revenues are defined inclusive of grants.

Table 1a. Tonga: External Debt Sustainability Framework, Baseline Scenario, 2010–33 1/

(In percent of GDP, unless otherwise indicated)

	Actual			Historical Average 6/	Standard Deviation 6/	Projections									
	2010	2011	2012			2013	2014	2015	2016	2017	2018	2013-2018 Average	2023	2033	2019-2033 Average
External debt (nominal) 1/	**31.8**	**35.4**	**41.2**			**40.3**	**38.9**	**37.1**	**35.1**	**33.2**	**31.3**		**23.0**	**14.2**	
of which: public and publicly guaranteed (PPG)	31.8	35.4	41.2			40.3	38.9	37.1	35.1	33.2	31.3		23.0	14.2	
Change in external debt	7.1	3.5	5.8			-0.9	-1.4	-1.8	-2.0	-1.9	-1.9		-1.5	-0.3	
Identified net debt-creating flows	-0.6	-2.8	1.9			3.4	1.8	0.7	0.1	-0.2	-0.3		0.0	-0.4	
Non-interest current account deficit	**4.3**	**2.2**	**6.1**	**3.9**	**3.6**	**5.1**	**4.0**	**3.0**	**2.5**	**2.2**	**2.2**		**2.6**	**2.6**	2.6
Deficit in balance of goods and services	35.5	32.6	36.3			26.7	26.9	28.4	29.2	29.6	29.5		29.1	29.3	
Exports	12.7	16.9	17.2			18.1	18.4	19.2	20.2	21.3	22.4		22.8	22.3	
Imports	48.2	49.5	53.5			44.8	45.2	47.6	49.4	50.9	51.9		51.9	51.6	
Net current transfers (negative = inflow)	-29.5	-27.8	-27.4	-31.3	2.8	-18.2	-19.4	-21.9	-23.2	-24.0	-23.9		-23.3	-23.6	-23.4
of which: official	-4.7	-4.8	-3.8			-1.0	-2.0	-4.7	-4.6	-4.3	-4.1		-2.7	-1.4	
Other current account flows (negative = net inflow)	-1.7	-2.7	-2.9			-3.5	-3.5	-3.5	-3.5	-3.5	-3.4		-3.3	-3.0	
Net FDI (negative = inflow)	**-2.0**	**-1.5**	**-1.1**	**-4.3**	**3.9**	**-2.3**	**-2.4**	**-2.4**	**-2.4**	**-2.5**	**-2.5**		**-2.6**	**-2.9**	-2.7
Endogenous debt dynamics 2/	**-2.8**	**-3.5**	**-3.1**			**0.6**	**0.2**	**0.1**	**0.0**	**0.0**	**0.0**		**0.0**	**-0.1**	
Contribution from nominal interest rate	0.6	0.6	0.5			0.8	0.7	0.7	0.7	0.6	0.6		0.4	0.1	
Contribution from real GDP growth	-0.7	-0.8	-0.3			-0.2	-0.6	-0.7	-0.6	-0.6	-0.6		-0.4	-0.2	
Contribution from price and exchange rate changes	-2.7	-3.2	-3.4			...	...	...	...	...	...		...	...	
Residual (3-4) 3/	**7.7**	**6.3**	**3.9**			**-4.2**	**-3.2**	**-2.4**	**-2.2**	**-1.7**	**-1.6**		**-1.5**	**0.1**	
of which: exceptional financing	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
PV of external debt 4/	...	...	35.7			35.3	33.8	32.0	29.9	27.9	26.0		17.4	9.2	
In percent of exports	...	...	207.5			195.1	184.1	166.9	147.9	131.0	115.9		76.5	41.4	
PV of PPG external debt	**...**	**...**	**35.7**			**35.3**	**33.8**	**32.0**	**29.9**	**27.9**	**26.0**		**17.4**	**9.2**	
In percent of exports	**...**	**...**	**207.5**			**195.1**	**184.1**	**166.9**	**147.9**	**131.0**	**115.9**		**76.5**	**41.4**	
In percent of government revenues	**...**	**...**	**197.6**			**194.8**	**186.8**	**174.4**	**161.9**	**148.9**	**137.6**		**89.8**	**46.7**	
Debt service-to-exports ratio (in percent)	**9.8**	**7.1**	**7.6**			**9.0**	**13.9**	**14.2**	**14.2**	**12.9**	**11.2**		**8.4**	**1.1**	
PPG debt service-to-exports ratio (in percent)	**9.8**	**7.1**	**7.6**			**9.0**	**13.9**	**14.2**	**14.2**	**12.9**	**11.2**		**8.4**	**1.1**	
PPG debt service-to-revenue ratio (in percent)	**6.2**	**6.1**	**7.2**			**8.9**	**14.1**	**14.9**	**15.5**	**14.6**	**13.3**		**9.9**	**1.3**	
Total gross financing need (Billions of U.S. dollars)	12.8	8.0	29.7			20.6	20.2	16.7	15.3	13.3	12.2		13.1	-0.2	
Non-interest current account deficit that stabilizes debt ratio	-2.8	-1.4	0.3			6.0	5.4	4.8	4.6	4.1	4.1		4.1	2.9	
Key macroeconomic assumptions															
Real GDP growth (in percent)	3.3	2.9	0.8	1.1	2.4	0.5	1.5	1.7	1.7	1.7	1.7	1.5	1.7	1.7	1.7
GDP deflator in US dollar terms (change in percent)	12.2	11.4	10.5	9.0	7.4	-0.9	1.8	1.8	1.9	2.0	2.2	1.5	2.4	2.4	2.4
Effective interest rate (percent) 5/	2.7	2.0	1.6	1.5	0.7	1.9	1.9	1.9	1.9	1.8	1.8	1.9	1.7	0.9	1.5
Growth of exports of G&S (US dollar terms, in percent)	11.3	52.0	13.6	10.4	19.3	4.6	5.0	8.1	9.3	9.3	9.3	7.6	4.0	4.0	4.2
Growth of imports of G&S (US dollar terms, in percent)	-1.6	17.8	20.5	12.8	12.4	-16.7	4.3	9.0	7.5	7.0	6.0	2.9	4.2	4.0	4.2
Grant element of new public sector borrowing (in percent)	...	...	...	...	...	11.1	35.4	35.4	35.4	35.4	35.4	31.3	35.4	35.4	35.4
Government revenues (excluding grants, in percent of GDP)	20.2	19.6	18.1			18.1	18.1	18.4	18.5	18.8	18.9		19.4	19.8	19.6
Aid flows (in Billions of US dollars) 7/	25.2	28.5	46.6			27.1	29.7	27.2	27.1	26.6	26.0		21.0	16.8	
of which: Grants	25.2	28.5	46.6			27.1	29.7	27.2	27.1	26.6	26.0		21.0	16.8	
of which: Concessional loans	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Grant-equivalent financing (in percent of GDP) 8/	...	...	...			5.9	6.6	5.9	5.7	5.4	5.1		3.4	1.8	2.9
Grant-equivalent financing (in percent of external financing) 8/	...	...	...			88.0	87.7	86.8	86.7	86.5	86.3		83.8	81.0	83.0
Memorandum items:															
Nominal GDP (Billions of US dollars)	369.2	423.0	471.6			469.5	485.1	502.3	520.6	540.2	561.9		690.6	1043.2	
Nominal dollar GDP growth	15.9	14.6	11.5			-0.4	3.3	3.5	3.6	3.8	4.0	3.0	4.2	4.2	4.2
PV of PPG external debt (in Billions of US dollars)			163.0			164.7	162.3	158.5	153.5	148.4	143.7		118.4	94.8	
(PVt-PVt-1)/GDPt-1 (in percent)						0.3	-0.5	-0.8	-1.0	-1.0	-0.9	-0.6	-0.8	0.2	-0.5
Gross workers' remittances (Billions of US dollars)	82.0	87.5	66.8			70.1	73.6	75.5	84.5	92.6	97.2		124.0	202.0	
PV of PPG external debt (in percent of GDP + remittances)	...	...	31.3			30.7	29.4	27.8	25.7	23.8	22.1		14.8	7.7	
PV of PPG external debt (in percent of exports + remittances)	...	...	113.8			106.8	100.8	93.6	82.0	72.6	65.4		42.8	22.2	
Debt service of PPG external debt (in percent of exports + remittances)	...	...	4.2			4.9	7.6	8.0	7.9	7.1	6.3		4.7	0.6	

Sources: Country authorities; and staff estimates and projections.

1/ Includes both public and private sector external debt.

2/ Derived as [r - g - ρ(1+g)]/(1+g+ρ+gρ) times previous period debt ratio, with r = nominal interest rate; g = real GDP growth rate, and ρ = growth rate of GDP deflator in U.S. dollar terms.

3/ Includes exceptional financing (i.e., changes in arrears and debt relief); changes in gross foreign assets; and valuation adjustments. For projections also includes contribution from price and exchange rate changes.

4/ Assumes that PV of private sector debt is equivalent to its face value.

5/ Current-year interest payments divided by previous period debt stock.

6/ Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.

7/ Defined as grants, concessional loans, and debt relief.

8/ Grant-equivalent financing includes grants provided directly to the government and through new borrowing (difference between the face value and the PV of new debt).

Table 1b. Tonga: Sensitivity Analysis for Key Indicators of Public and Publicly Guaranteed External Debt, 2013–33 1/

(In percent)

	Projections							
	2013	2014	2015	2016	2017	2018	**2023**	2033
PV of debt-to GDP ratio								
Baseline	35	34	32	30	28	26	**17**	9
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2013-2033 1/	35	30	26	22	19	16	**6**	-1
A2. New public sector loans on less favorable terms in 2013-2033 2	35	34	33	31	29	28	**21**	15
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2014-2015	35	34	33	31	29	27	**18**	10
B2. Export value growth at historical average minus one standard deviation in 2014-2015 3/	35	35	37	35	33	31	**21**	11
B3. US dollar GDP deflator at historical average minus one standard deviation in 2014-2015	35	34	32	30	28	26	**17**	9
B4. Net non-debt creating flows at historical average minus one standard deviation in 2014-2015 4/	35	29	24	22	20	18	**11**	6
B5. Combination of B1-B4 using one-half standard deviation shocks	35	26	19	17	15	14	**8**	4
B6. One-time 30 percent nominal depreciation relative to the baseline in 2014 5/	35	47	45	42	39	36	**24**	13
PV of debt-to-exports ratio								
Baseline	195	184	167	148	131	116	**77**	41
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2013-2033 1/	195	163	134	109	89	73	**28**	-4
A2. New public sector loans on less favorable terms in 2013-2033 2	195	186	170	153	138	125	**93**	68
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2014-2015	195	182	165	146	129	114	**75**	40
B2. Export value growth at historical average minus one standard deviation in 2014-2015 3/	195	221	262	234	209	187	**127**	69
B3. US dollar GDP deflator at historical average minus one standard deviation in 2014-2015	195	182	165	146	129	114	**75**	40
B4. Net non-debt creating flows at historical average minus one standard deviation in 2014-2015 4/	195	156	124	107	93	80	**48**	26
B5. Combination of B1-B4 using one-half standard deviation shocks	195	150	116	99	84	71	**39**	21
B6. One-time 30 percent nominal depreciation relative to the baseline in 2014 5/	195	182	165	146	129	114	**75**	40
PV of debt-to-revenue ratio								
Baseline	195	187	174	162	149	138	**90**	47
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2013-2033 1/	195	166	140	119	101	86	**32**	-4
A2. New public sector loans on less favorable terms in 2013-2033 2	195	188	178	168	157	148	**109**	77
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2014-2015	195	190	182	169	155	144	**93**	48
B2. Export value growth at historical average minus one standard deviation in 2014-2015 3/	195	194	200	187	174	162	**109**	57
B3. US dollar GDP deflator at historical average minus one standard deviation in 2014-2015	195	185	172	160	147	136	**88**	46
B4. Net non-debt creating flows at historical average minus one standard deviation in 2014-2015 4/	195	158	129	117	105	95	**57**	29
B5. Combination of B1-B4 using one-half standard deviation shocks	195	143	104	93	82	73	**40**	21
B6. One-time 30 percent nominal depreciation relative to the baseline in 2014 5/	195	262	244	226	207	192	**125**	65

Table 1b. Tonga: Sensitivity Analysis for Key Indicators of Public and Publicly Guaranteed External Debt, 2013–33 (continued)

(In percent)

Debt service-to-exports ratio								
Baseline	9	14	14	14	13	11	**8**	1
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2013-2033 1/	9	12	12	11	9	8	**4**	0
A2. New public sector loans on less favorable terms in 2013-2033 2	9	13	14	14	13	11	**9**	4
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2014-2015	9	13	14	14	12	11	**9**	2
B2. Export value growth at historical average minus one standard deviation in 2014-2015 3/	9	15	19	19	17	15	**13**	4
B3. US dollar GDP deflator at historical average minus one standard deviation in 2014-2015	9	13	14	14	12	11	**9**	2
B4. Net non-debt creating flows at historical average minus one standard deviation in 2014-2015 4/	9	13	13	13	12	10	**7**	1
B5. Combination of B1-B4 using one-half standard deviation shocks	9	14	15	14	13	11	**7**	1
B6. One-time 30 percent nominal depreciation relative to the baseline in 2014 5/	9	13	14	14	12	11	**9**	2
Debt service-to-revenue ratio								
Baseline	9	14	15	16	15	13	**10**	1
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2013-2033 1/	9	13	12	12	11	9	**5**	0
A2. New public sector loans on less favorable terms in 2013-2033 2	9	13	14	15	14	13	**11**	4
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2014-2015	9	14	15	16	15	14	**11**	3
B2. Export value growth at historical average minus one standard deviation in 2014-2015 3/	9	13	14	15	14	13	**11**	3
B3. US dollar GDP deflator at historical average minus one standard deviation in 2014-2015	9	14	14	15	14	13	**10**	3
B4. Net non-debt creating flows at historical average minus one standard deviation in 2014-2015 4/	9	13	14	14	13	12	**8**	1
B5. Combination of B1-B4 using one-half standard deviation shocks	9	13	13	14	13	12	**7**	1
B6. One-time 30 percent nominal depreciation relative to the baseline in 2014 5/	9	19	20	21	20	18	**14**	4
Memorandum item:								
Grant element assumed on residual financing (i.e., financing required above baseline) 6/	31	31	31	31	31	31	**31**	31

Sources: Country authorities; and staff estimates and projections.

1/ Variables include real GDP growth, growth of GDP deflator (in U.S. dollar terms), non-interest current account in percent of GDP, and non-debt creating flows.

2/ Assumes that the interest rate on new borrowing is by 2 percentage points higher than in the baseline., while grace and maturity periods are the same as in the baseline.

3/ Exports values are assumed to remain permanently at the lower level, but the current account as a share of GDP is assumed to return to its baseline level after the shock (implicitly assuming an offsetting adjustment in import levels).

4/ Includes official and private transfers and FDI.

5/ Depreciation is defined as percentage decline in dollar/local currency rate, such that it never exceeds 100 percent.

6/ Applies to all stress scenarios except for A2 (less favorable financing) in which the terms on all new financing are as specified in footnote 2.

Table 2a. Tonga: Public Sector Debt Sustainability Framework, Baseline Scenario, 2010–33 1/

(In percent of GDP, unless otherwise indicated)

	Actual					Estimate						Projections			
	2010	2011	2012	Average 5/	Standard Deviation 5/	2013	2014	2015	2016	2017	2018	2013-18 Average	2023	2033	2019-33 Average
Public sector debt 1/	40.0	40.7	45.8			44.6	43.0	41.0	38.7	36.5	34.4		25.1	15.2	
of which: foreign-currency denominated	31.8	35.4	41.2			40.3	38.9	37.1	35.1	33.2	31.3		23.0	14.2	
Change in public sector debt	7.7	0.7	5.1			-1.2	-1.6	-2.0	-2.3	-2.2	-2.1		-1.7	-0.4	
Identified debt-creating flows	2.5	0.5	2.2			-1.4	-1.2	-2.7	-3.3	-3.2	-2.4		-1.5	-0.9	
Primary deficit	4.5	6.8	1.9	0.9	3.1	-1.2	-0.9	-2.2	-2.7	-2.5	-1.6	-1.9	-0.8	-0.4	-0.8
Revenue and grants	27.0	26.3	28.0			23.9	24.2	23.8	23.7	23.7	23.5		22.4	21.4	
of which: grants	6.8	6.7	9.9			5.8	6.1	5.4	5.2	4.9	4.6		3.0	1.6	
Primary (noninterest) expenditure	31.5	33.1	29.9			22.6	23.3	21.6	21.0	21.1	21.8		21.6	21.0	
Automatic debt dynamics	-2.0	-6.4	0.8			0.0	-0.3	-0.5	-0.6	-0.6	-0.8		-0.6	-0.5	
Contribution from interest rate/growth differential	-0.7	-1.2	-0.2			0.4	-0.1	-0.4	-0.4	-0.4	-0.4		-0.3	-0.3	
of which: contribution from average real interest rate	0.4	0.0	0.1			0.6	0.6	0.4	0.3	0.3	0.3		0.1	0.0	
of which: contribution from real GDP growth	-1.0	-1.1	-0.3			-0.2	-0.7	-0.7	-0.7	-0.7	-0.6		-0.5	-0.3	
Contribution from real exchange rate depreciation	-1.4	-5.2	1.0			-0.4	-0.2	-0.1	-0.3	-0.3	-0.4		...	...	
Other identified debt-creating flows	0.0	0.1	-0.6			-0.2	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Privatization receipts (negative)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Recognition of implicit or contingent liabilities	0.0	0.1	-0.6			-0.2	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Debt relief (HIPC and other)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Other (specify, e.g. bank recapitalization)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Residual, including asset changes	5.3	0.2	2.9			0.3	-0.4	0.7	1.0	1.0	0.3		-0.2	0.5	
Other Sustainability Indicators															
PV of public sector debt	...	...	40.3			39.6	37.9	35.9	33.5	31.3	29.1		19.6	10.3	
of which: foreign-currency denominated	...	...	35.7			35.3	33.8	32.0	29.9	27.9	26.0		17.4	9.2	
of which: external	...	...	35.7			35.3	33.8	32.0	29.9	27.9	26.0		17.4	9.2	
PV of contingent liabilities (not included in public sector debt)	...	...	...			...	...	...	...	...	...		...	...	
Gross financing need 2/	6.0	9.6	3.8			1.5	2.6	1.5	1.0	0.7	1.7		1.7	0.1	
PV of public sector debt-to-revenue and grants ratio (in percent)	...	...	144.2			165.8	156.4	150.9	141.5	132.1	123.9		87.3	47.9	
PV of public sector debt-to-revenue ratio (in percent)	...	...	223.1			218.5	209.3	195.4	181.4	166.8	154.3		100.9	51.8	
of which: external 3/	...	...	197.6			194.8	186.8	174.4	161.9	148.9	137.6		89.8	46.7	
Debt service-to-revenue and grants ratio (in percent) 4/	5.5	10.7	6.8			11.7	14.4	15.5	15.8	13.6	14.3		11.2	2.4	
Debt service-to-revenue ratio (in percent) 4/	7.4	14.4	10.5			15.4	19.3	20.1	20.2	17.2	17.8		12.9	2.6	
Primary deficit that stabilizes the debt-to-GDP ratio	-3.2	6.1	-3.1			-0.1	0.7	-0.2	-0.4	-0.4	0.5		0.8	0.0	
Key macroeconomic and fiscal assumptions															
Real GDP growth (in percent)	3.3	2.9	0.8	1.1	2.4	0.5	1.5	1.7	1.7	1.7	1.7	1.5	1.7	1.7	1.7
Average nominal interest rate on forex debt (in percent)	2.7	2.0	1.6	1.5	0.7	1.9	1.9	1.9	1.9	1.8	1.8	1.9	1.7	0.9	1.5
Average real interest rate on domestic debt (in percent)	-0.3	-2.0	1.8	-3.1	4.6	2.8	2.4	1.6	1.1	0.7	0.5	1.5	0.0	0.0	0.1
Real exchange rate depreciation (in percent, + indicates depreciation	-5.6	-16.8	3.0	-5.3	8.2	-1.0	...	...	...	...	...	...	...	...	...
Inflation rate (GDP deflator, in percent)	3.7	5.8	2.3	6.2	5.0	2.9	3.5	4.4	4.9	5.3	5.5	4.4	6.0	6.0	6.0
Growth of real primary spending (deflated by GDP deflator, in percer	0.0	0.1	-0.1	0.0	0.2	-0.2	0.0	-0.1	0.0	0.0	0.1	0.0	0.0	0.0	0.0
Grant element of new external borrowing (in percent)	...	...	...	...	...	11.1	35.4	35.4	35.4	35.4	35.4	31.3	35.4	35.4	...

Sources: Country authorities; and staff estimates and projections.

1/ [Indicate coverage of public sector, e.g., general government or nonfinancial public sector. Also whether net or gross debt is used.]

2/ Gross financing need is defined as the primary deficit plus debt service plus the stock of short-term debt at the end of the last period.

3/ Revenues excluding grants.

4/ Debt service is defined as the sum of interest and amortization of medium and long-term debt.

5/ Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.

Table 2b. Tonga: Sensitivity Analysis for Key Indicators of Public Debt, 2013–33 1/

	Projections							
	2013	2014	2015	2016	2017	2018	2023	2033
PV of Debt-to-GDP Ratio								
Baseline	40	38	36	34	31	29	20	10
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	40	39	40	40	41	41	38	36
A2. Primary balance is unchanged from 2013	40	38	36	35	34	32	22	7
A3. Permanently lower GDP growth 1/	40	38	36	34	33	31	25	26
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2014-2015	40	39	39	38	36	34	28	25
B2. Primary balance is at historical average minus one standard deviations in 2014-2015	40	42	44	42	39	37	26	16
B3. Combination of B1-B2 using one half standard deviation shocks	40	41	43	41	39	37	29	22
B4. One-time 30 percent real depreciation in 2014	40	53	51	48	45	42	31	19
B5. 10 percent of GDP increase in other debt-creating flows in 2014	40	45	43	41	38	36	26	15
PV of Debt-to-Revenue Ratio 2/								
Baseline	166	156	151	142	132	124	87	48
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	166	162	168	170	172	172	169	168
A2. Primary balance is unchanged from 2013	166	155	153	148	143	136	97	33
A3. Permanently lower GDP growth 1/	166	157	153	145	137	131	109	121
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2014-2015	166	161	163	157	150	145	124	114
B2. Primary balance is at historical average minus one standard deviations in 2014-2015	166	172	186	176	166	157	118	73
B3. Combination of B1-B2 using one half standard deviation shocks	166	169	180	173	164	158	129	102
B4. One-time 30 percent real depreciation in 2014	166	220	214	202	191	180	136	89
B5. 10 percent of GDP increase in other debt-creating flows in 2014	166	187	182	173	163	154	115	70
Debt Service-to-Revenue Ratio 2/								
Baseline	12	14	15	16	14	14	11	2
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	12	14	16	16	15	17	16	9
A2. Primary balance is unchanged from 2013	12	14	15	16	14	15	12	2
A3. Permanently lower GDP growth 1/	12	14	16	16	14	15	12	6
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2014-2015	12	15	16	17	15	16	13	6
B2. Primary balance is at historical average minus one standard deviations in 2014-2015	12	14	16	17	16	17	13	4
B3. Combination of B1-B2 using one half standard deviation shocks	12	15	16	17	16	17	14	6
B4. One-time 30 percent real depreciation in 2014	12	17	21	21	19	20	17	5
B5. 10 percent of GDP increase in other debt-creating flows in 2014	12	14	16	17	16	17	13	4

Sources: Country authorities; and staff estimates and projections.

1/ Assumes that real GDP growth is at baseline minus one standard deviation divided by the square root of the length of the projection period.

2/ Revenues are defined inclusive of grants.